

SMART ICU ASSISTANT: A MACHINE LEARNING PIPELINE FOR REAL-TIME ICU PATIENT MONITORING, CLINICAL RISK PREDICTION, AND EARLY WARNING

Submitted by
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In Partial fulfillment of the requirements for the award of
Master of Science in Computer Science with Specialization in Data Analytics
of



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May 2026

BONAFIDE CERTIFICATE

This is to certify that the project report entitled **“Smart ICU Assistant — A Machine Learning Pipeline for Real-Time ICU Patient Monitoring, Clinical Risk Prediction, and Early Warning”** submitted by **Fadhil Muhammed N C** (Reg. No: 243014) in partial fulfillment of the requirements for the award of **Master of Science in Computer Science with Specialization in Data Analytics** is a bonafide record of the work carried out at **Kerala University of Digital Sciences, Innovation and Technology** under my supervision. This work has not been submitted elsewhere for the award of any other degree or diploma.

Supervisor

Dr. Manoj Kumar T K
Dean(HR) and Professor
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CERTIFICATE

This is to certify that the report titled Smart ICU Assistant is based on the work carried out by Fadhil Muhammed N C, 243014 during the period of his study at the School of Digital Sciences, Kerala University of Digital Sciences, Innovation and Technology (Digital University of Kerala) under the supervision of Manoj Kumar T K, Dean (HR) and Professor.

This report is submitted to the Kerala University of Digital Sciences, Innovation and Technology, in partial fulfilment of the requirements for the award of Master of Science in Computer Science with Specialization in Data Analytics. No part of this report has been submitted earlier for the award of any other degree.

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Declaration

I, **Fadhil Muhammed N C**, a student of Master of Science in Computer Science with specialization in Data Analytics at Digital University Kerala, hereby declare that this project report entitled “**Smart ICU Assistant — A Machine Learning Pipeline for Real-Time ICU Patient Monitoring, Clinical Risk Prediction, and Early Warning**” is the result of my own work carried out during the period January 2026 to May 2026. This report has not been submitted elsewhere for the award of any degree or diploma. All sources of information and references have been duly acknowledged.



Fadhil Muhammed N C

Place: Thiruvananthapuram

Date: 15 / 05 / 25

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And my family and friends—thanks for their constant support and inspiration..

Fadhil Muhammed N C

Abstract

Intensive Care Units generate approximately 1,500 data points per patient per day, yet the vast majority of this information is not utilised in real-time clinical decision making. This report presents the Smart ICU Assistant, a modular machine learning pipeline that processes 43.6 GB of clinical records from the MIMIC-III database (61,532 ICU stays) to predict six clinical outcomes simultaneously: ICU mortality (6, 12, 24-hour horizons), sepsis onset using SIRS criteria (6, 12, 24 h), acute kidney injury at KDIGO stages 1–3 (24, 48 h), vasopressor requirement (6, 12 h), mechanical ventilation need (6, 12, 24 h), and ICU length of stay (short/long classification). This yields a total of 19 binary prediction labels.

The feature engineering pipeline extracts 81 features per timestep from 8 vital signs and 7 laboratory values, including derived clinical indices (shock index, pulse pressure, SIRS score) and rolling statistics at three window sizes. Four model architectures are trained per task: BiLSTM with temporal attention, Pre-LayerNorm Transformer, XGBoost (with class weighting), and LightGBM. Model outputs are combined through AUROC²-weighted averaging and a stacking meta-learner, with per-task threshold tuning to maximise F1 score.

Experimental results demonstrate adequate-to-good discriminative performance: ventilation 0.85 AUROC (strongest result), vasopressor 0.80/0.81, mortality 0.77/0.78, sepsis 0.76/0.77, and AKI Stage 1 0.85. Tree-based models—particularly LightGBM—consistently outperformed deep learning architectures across all tasks, likely attributable to the threshold-based nature of clinical criteria and the limited sequence length (24 timesteps). All training was conducted in FP16 mixed precision on consumer-grade hardware (RTX 3050, 4 GB VRAM).

Keywords: Intensive Care Unit, Clinical Decision Support, MIMIC-III, Deep Learning, LSTM, Transformer, XGBoost, LightGBM, Ensemble Learning, Sepsis Prediction, Acute Kidney Injury, Mortality Prediction, SHAP, Mixed-Precision Training, FastAPI

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Chapter 1

Introduction

1.1 Introduction

On average, a single Intensive Care Unit (ICU) produces 1,500 points of data per patient daily, including physio- logical parameters, lab tests, drug treatments, and other clinical findings. In spite of the wealth of data being produced, however, signs of clinical deterioration often go unnoticed until a point of serious decline is reached, due to the inability of clinicians dealing with a number of patients at once to monitor every single trend in real-time. This is where machine learning (ML) proves itself as a feasible tool of assistance in the process: a computer-based system able to monitor patient's state on a constant basis and alert the medical personnel to any signs of impending risk.

This report presents the **Smart ICU Assistant**, an integrated clinical prediction system that employs machine learning techniques to forecast patient deterioration in ICU settings. The system processes 43.6 GB of clinical monitoring data from the open-source MIMIC-III database [1], predicting six distinct clinical outcomes simultaneously through 19 individual binary prediction labels. Four model architectures are trained per task and combined using ensemble methods to maximise predictive performance.

1.2 Background and Problem Statement

For decades, traditional ICU scoring systems, such as the APACHE-II [2] and SOFA [3] have been the standard for the clinical world. These instruments, however, have some basic limitations: APACHE-II is calculated only once at the time of admission and never updated thereafter; SOFA is updated daily only. In addition, both systems use linear scoring functions, which do not naturally support non-linear interactions between physiological variables that are often present during early stages of deterioration, such as HRV and blood pressure during the prodromal period of sepsis [4].

MIMIC-III [1] was a huge step forward for data-driven clinical research: 61,532 ICU visits, 26 relational tables, and 43.6 GB of de-identified clinical data from a single tertiary hospital. Harutyunyan et al. [5] established the primary benchmark on this dataset—four clinical tasks, a single-architecture LSTM, achieving AUROC 0.870 for mortality prediction. While this work demonstrated the feasibility of ML-based clinical prediction, it was limited in scope: four tasks, one model architecture, no ensemble methods, and no model interpretability.

1.2.1 Problem Formulation

With regard to physiological vital signs as well as lab measurements for patients admitted to the Intensive Care Unit, the problem tackled by this project can be formulated as follows:

The first step is the design, the second step is the implementation. The last step is the evaluation of a machine learning pathway that is modularized. The pipeline is designed to process multivariate temporal ICU data from MIMICIII, and produce calibrated probability estimates for 19 binary clinical event labels grouped into 6 categories.

1.2.2 Clinical Prediction Tasks

The system addresses six clinically significant prediction tasks:

1. **ICU Mortality** — Prediction of patient death within 6, 12, or 24 hours (3 labels).
2. **Sepsis Onset** — Detection of developing sepsis ($\text{SIRS} \geq 2$ with infection evidence) within 6, 12, or 24 hours (3 labels).
3. **Acute Kidney Injury** — Prediction of AKI at KDIGO stages 1–3 within 24 or 48 hours (6 labels).
4. **Vasopressor Requirement** — Prediction of vasopressor initiation within 6 or 12 hours (2 labels).
5. **Mechanical Ventilation** — Prediction of ventilator initiation within 6, 12, or 24 hours (3 labels).
6. **Length of Stay** — Binary classification of short stay ($<24\text{h}$ remaining) or long stay ($>72\text{h}$ remaining) (2 labels).

1.3 Motivation

The disparity between the static nature of existing scoring systems and the dynamic capabilities of contemporary ML methods motivated this project. The following nine objectives were defined:

- O1:** Develop a data ingestion pipeline capable of processing 43.6 GB of MIMIC-III data on hardware with limited memory, employing chunked reading of CHARTEVENTS with item-level filtering.
- O2:** Extract 81 features per timestep from 8 vital signs and 7 laboratory values, incorporating rolling statistics and derived clinical indices such as shock index.
- O3:** Implement label generators for all six tasks (19 labels) using clinically validated definitions—KDIGO criteria for AKI, SIRS criteria for sepsis, and temporal event detection for the remaining tasks.
- O4:** Train four model architectures per task: BiLSTM-Attention, Pre-LayerNorm Transformer, XGBoost, and LightGBM, giving both sequential and tabular coverage.
- O5:** Combine predictions via AUROC²-weighted averaging and stacking to evaluate whether ensemble methods provide consistent improvement.
- O6:** Tune classification thresholds per task on validation F1, as the default threshold of 0.5 is inappropriate when positive prevalence is below 5%.
- O7:** Incorporate SHAP-based feature importance for the tree-based models to support clinical interpretability.
- O8:** Build a FastAPI dashboard providing real-time monitoring, composite risk scores and clinical alerts.
- O9:** Benchmark performance against published baselines and document both successful approaches and engineering challenges encountered during development.

1.4 Scope of the Project

In Scope

- Retrospective development and evaluation on MIMIC-III.
- Six clinical tasks, 19 binary labels.
- Four model architectures per task.
- Ensemble methods (weighted averaging and stacking).
- Web dashboard for visualisation and clinical alerts.
- SHAP-based interpretability for tree-based models.
- FP16 mixed-precision training on a consumer GPU.

Out of Scope

- Live patient data or electronic health record integration.
- Regulatory approval of any kind.
- Unstructured data sources (clinical notes, radiology, waveforms).
- External validation on eICU, MIMIC-IV, HiRID, or other datasets.
- Clinical deployment in a hospital setting.

1.5 Organization of the Report

The literature review in Chapter 2 will provide an overview of the existing literature and highlight the gaps in the literature that this work will help fill. The data pipeline, with loading strategies and feature engineering are presented in Chapter 3. The methodology is detailed in Chapter 4 under the topics of label generation, model architectures, ensemble construction and system design. Chapter 5 reports experimental results and analysis. Chapter 6 discusses future directions. A summary is given in Chapter 7.

Chapter 2

Literature Review

2.1 Introduction

This chapter reviews current literature on clinical risk prediction in the ICU, which includes traditional scoring systems, deep-learning methods, gradient-boosted tree methods, clinical prediction research specific to the task, and the interpretability of models. The gap analysis brings together the deficiencies of previous research and situates the work of this project.

2.2 State-of-the-art Approaches

2.2.1 Clinical Risk Scoring in the ICU

APACHE-II [2] was among the earliest severity-of-illness scoring systems: 12 physiological variables from the first 24 hours yield a single severity score. SOFA [3] tracks organ dysfunction across six organ systems, refreshing daily. qSOFA [4] provides a rapid bedside assessment using three criteria, while the National Early Warning Score (NEWS) aggregates six vital sign parameters. Although widely adopted, these instruments share a fundamental limitation: they employ linear additive scoring, which cannot capture the non-linear interactions between physiological variables—such as the relationship between heart rate variability and blood pressure—that often characterise evolving clinical deterioration.

2.2.2 Recurrent Neural Networks and Attention

The foundational work on the use of multi-tasking in clinical predictions in MIMIC-III was done by Harutyunyan et al. [5] who solved four clinical prediction problems (mortality prediction, decompensation, length of stay prediction, and phenotyping), using only one multi-task LSTM network, achieving AUROC 0.870 for

mortality prediction. Bidirectional LSTMs were then used to tackle the problem from two directions, essentially doubling the amount of context that the model learned at each step. The use of temporal attention networks further improved upon this method through assigning importance scores to the particular steps, allowing clinicians to understand more precisely the relevant time frames.

2.2.3 Transformers in Clinical Data

The Transformer architecture [6] replaces recurrence with multi-head self-attention, enabling parallel processing and direct modelling of dependencies between distant time points. However, numerical stability presents a practical challenge: under FP16 mixed-precision training, the original Post-LayerNorm formulation can produce attention scores that overflow the representable range, resulting in NaN losses. The Pre-LayerNorm variant [7]—which applies layer normalisation *before* each sublayer rather than after—resolves this instability. constraints.

2.2.4 Gradient-Boosted Trees

When it comes to structured clinical data, the gradient boosted tree ensembles are still competitive with deep learning approaches. XGBoost [8] has native support for missing values without imputation, class weighting at the instance level for imbalanced data and provides GPU accelerated histogram splitting. LightGBM [9] uses leaf-wise growth (which means different from level-wise) and Gradient-based One-Side Sampling (GOSS), generating structurally different tree ensembles based on the same set of features. This is exploited in the form of both frameworks to increase the robustness of ensemble predictions.

2.2.5 Task-Specific Literature

Sepsis is defined as a dysregulated host response to infection leading to organ dysfunction [4]—is associated with substantial mortality that increases with each hour of delayed treatment. The PhysioNet 2019 Challenge [10] specifically targeted early sepsis prediction, with top-performing models achieving AUROCs exceeding 0.85.

Acute Kidney Injury (AKI) is diagnosed via the KDIGO creatinine criteria [11] and affects 20–50% of ICU patients [12]. Tomasev et al. [13] achieved AUROC 0.921 for 48-hour AKI prediction; however, this result was obtained on proprietary Veterans Affairs data and has not been replicated on a publicly available dataset.

2.2.6 Interpretability in Clinical AI

Clinical adoption of predictive models requires interpretability: clinicians are unlikely to trust predictions from opaque models. SHAP [14] provides theoretically grounded feature importance scores based on Shapley

values from cooperative game theory, allocating credit (or blame) to each feature proportional to its marginal contribution.

2.3 Summary

2.3.1 Gap Analysis

Five principal gaps in the existing literature are as follows:

1. Limited task coverage: The primary benchmark [5] addresses four clinical tasks. This work extends coverage to six tasks (19 binary labels).
2. Single architecture: Previous work typically employs a single model type. This project trains four architectures per task and evaluates their strength.
3. Absence of ensemble methods and threshold optimisation AUROC²-weighted averaging, stacking, and per-task F1-optimal threshold tuning are introduced.
4. No interpretability : SHAP-based feature importance is incorporated for the tree-based models.

Table 2.1 summarises the comparison with selected prior work.

Table 2.1: Comparison with selected prior work on MIMIC-III clinical prediction.

Property	Harutyunyan (2019)	Tomasev (2019)	Moor (2021)	This Work
Dataset	MIMIC-III	VA EHR	MIMIC-III	MIMIC-III
Clinical tasks	4	1 (AKI)	1 (Sepsis)	6
Prediction labels	4	3	1	19
Model types	1	1	1	4
Ensemble	No	No	No	Yes
Threshold tuning	No	No	No	Yes
SHAP	No	No	No	Yes
Web dashboard	No	No	No	Yes

Chapter 3

Data Pipeline and Feature Engineering

3.1 Introduction

This chapter describes the data acquisition, preprocessing, and feature engineering pipeline underpinning the Smart ICU Assistant. It covers the source database, the strategy for loading 43.6 GB of clinical records into memory, the merging of 17 relational tables, the exploratory data analysis (EDA) findings, and the construction of the 81-feature temporal matrices used as model inputs.

3.2 The MIMIC-III Clinical Database

MIMIC-III [1] is a publicly available, de-identified clinical database comprising ICU records from Beth Israel Deaconess Medical Center (BIDMC), spanning 2001–2012. The dataset is distributed through PhysioNet [15] and requires completion of CITI human subjects training and execution of a data use agreement. All records are HIPAA Safe Harbor compliant: dates are shifted, direct identifiers are removed, and patient ages exceeding 89 years are replaced with the sentinel value of 300. The dataset was used exclusively for academic research purposes.

In terms of scale, MIMIC-III encompasses 46,520 unique patients, 58,976 hospital admissions, 61,532 ICU stays, and 26 relational tables, of which 17 are utilised in this pipeline. The total uncompressed size is 43.6 GB. The CHARTEVENTS table alone accounts for 33.6 GB, comprising approximately 330 million rows.

3.3 Dataset Tables and Loading Strategy

Table 3.1 enumerates the MIMIC-III tables incorporated into the pipeline and their respective roles.

Table 3.1: MIMIC-III tables used in the Smart ICU Assistant pipeline.

Table	Approx. Size	Role in Pipeline
PATIENTS	2 MB	Demographics, date of birth/death
ADMISSIONS	8 MB	Admission/discharge times, admission type
ICUSTAYS	3 MB	ICU stay times, care unit, length of stay
CHARTEVENTS	33.6 GB	Vital signs (HR, BP, SpO ₂ , RR, temp, glucose)
LABEVENTS	2.3 GB	Laboratory results (creatinine, lactate, WBC, etc.)
D_ITEMS	0.5 MB	Dictionary for CHARTEVENTS item IDs
D_LABITEMS	0.1 MB	Dictionary for LABEVENTS item IDs
PRESCRIPTIONS	500 MB	Medication prescriptions (antibiotics, vasopressors)
INPUTEVENTS_MV	200 MB	IV inputs (MetaVision, vasopressor administration)
OUTPUTEVENTS	150 MB	Patient outputs (urine output)
DIAGNOSES_ICD	15 MB	ICD-9 diagnosis codes (sepsis, infection)
PROCEDURES_ICD	5 MB	ICD-9 procedure codes (ventilation)
SERVICES	3 MB	Service transfers
TRANSFERS	5 MB	Physical location transfers
CALLOUT	2 MB	Discharge readiness records
DATETIMEEVENTS	100 MB	Date/time-stamped events
MICROBIOLOGYEVENTS	30 MB	Microbiology culture results

A two-phase loading strategy was adopted. Tables smaller than 100 MB were loaded in their entirety using `pandas.read_csv()`. Larger tables—specifically CHARTEVENTS and LABEVENTS—required specialised chunked processing, as described below.

3.3.1 Chunked Loading of CHARTEVENTS

The CHARTEVENTS table contains approximately 330 million rows (33.6 GB), rendering full in-memory loading infeasible on the available hardware (16 GB RAM), as naïve loading would require over 40 GB of memory. To mitigate this problem, we processed the table bit, by bit by loading it in chunks of two million rows at a time. Each chunk was filtered immediately upon reading, keeping only rows whose item IDs matched the target vital signs. This approach reduced memory usage by approximately 85%.

3.3.2 LABEVENTS: ICU Stay Time-Overlap Join

The LABEVENTS table contains 27 million rows of laboratory results. However, lab records are keyed by `SUBJECT_ID` and `HADM_ID` (hospital admission), not by ICU stay. Consequently, a given creatinine measurement has no direct link to a specific ICU stay. This was resolved through a time-based join: each laboratory result's `CHARTTIME` was compared against the `INTIME/OUTTIME` windows of all ICU stays for the corresponding patient, and the result was assigned to the temporally overlapping stay.

3.4 Dataset Statistics

Table 3.2 presents the summary statistics of the processed dataset.

Table 3.2: Summary statistics of the processed MIMIC-III dataset.

Statistic	Value
Total dataset size (uncompressed)	43.6 GB
Largest single table (CHARTEVENTS)	33.6 GB
Unique patients	46,520
Hospital admissions	58,976
ICU stays	61,532
Median age at admission (years)	65.8
Male patients (%)	55.9
In-hospital mortality rate (%)	11.5
Median ICU length of stay (hours)	50.4
CHARTEVENTS rows after filtering	~42 million
LABEVENTS rows after time-overlap join	~13 million
Total features per timestep	81
Prediction labels	19

3.5 Data Merging and Derived Fields

All 17 tables were merged into a single longitudinal record per ICU stay, following this sequence:

1. PATIENTS + ADMISSIONS on `SUBJECT_ID`: demographic and admission information.
2. + ICUSTAYS on `HADM_ID`: ICU timing and care unit assignment.

3. + CHARTEVENTS (filtered) on ICUSTAY_ID + CHARTTIME: vital signs.
4. + LABEVENTS via time-overlap join: laboratory results.
5. + PRESCRIPTIONS + INPUTEVENTS_MV: medication records.
6. + DIAGNOSES_ICD + PROCEDURES_ICD: diagnostic and procedural codes for label generation.

3.6 Exploratory Data Analysis

Exploratory data analysis (EDA) was used to describe the statistical characteristics, distribution and clinical structure of the data set before developing the model.

3.6.1 Patient Age Distribution

Figure 3.1 presents the age distribution stratified by survival outcome. Mortality rates exceed 15% among patients aged over 80.

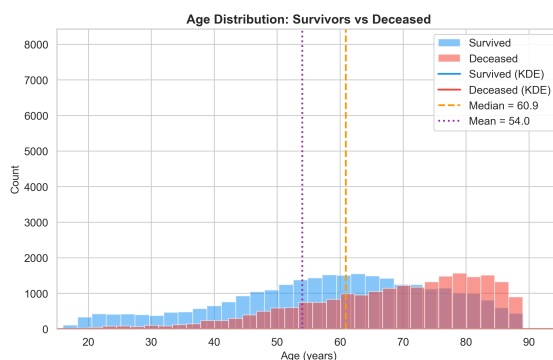


Figure 3.1: Age distribution by survival outcome.

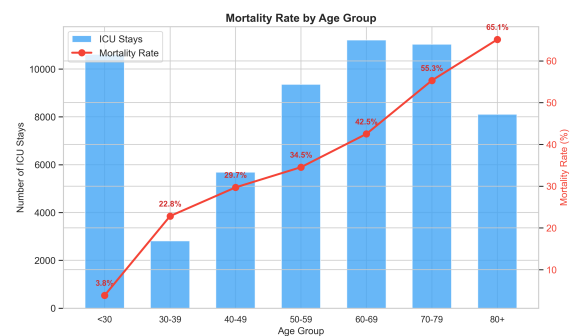


Figure 3.2: Mortality rate by age group.

55.9% of the cohort are male patients; the most common admission type is emergency, which is associated with higher mortality. The Medical ICU (MICU) has the highest volume of admissions and the highest mortality rate among care units. The distribution of ICU length of stay is heavily right-skewed with a median of approximately 50 hours, but a long right tail extending beyond 3,000 hours.

Vital sign box plots indicate frequent triggering of SIRS criteria: the median heart rate is 92 bpm (above the 90 bpm SIRS threshold), and temperature values show wide dispersion around the febrile cutoff. Laboratory distributions reveal that creatinine values exhibit wide dispersion above the 1.2 mg/dL KDIGO threshold, while WBC and lactate show extreme right-skew due to outlier values in septic patients.

3.6.2 Feature Correlation

The inter-feature correlation matrix (Figure 3.3) verifies some physiological correlations between the three blood pressure features, with correlation values > 0.7 . This is expected due to the high correlation between SBP and DBP and between MAP and DBP. Also, a moderate correlation exists between the laboratory values, as seen in the cross-correlation between LDL and HDL, or between HDL and HDL. This intercorrelation is beneficial to tree-based splitting and neural attention mechanisms.

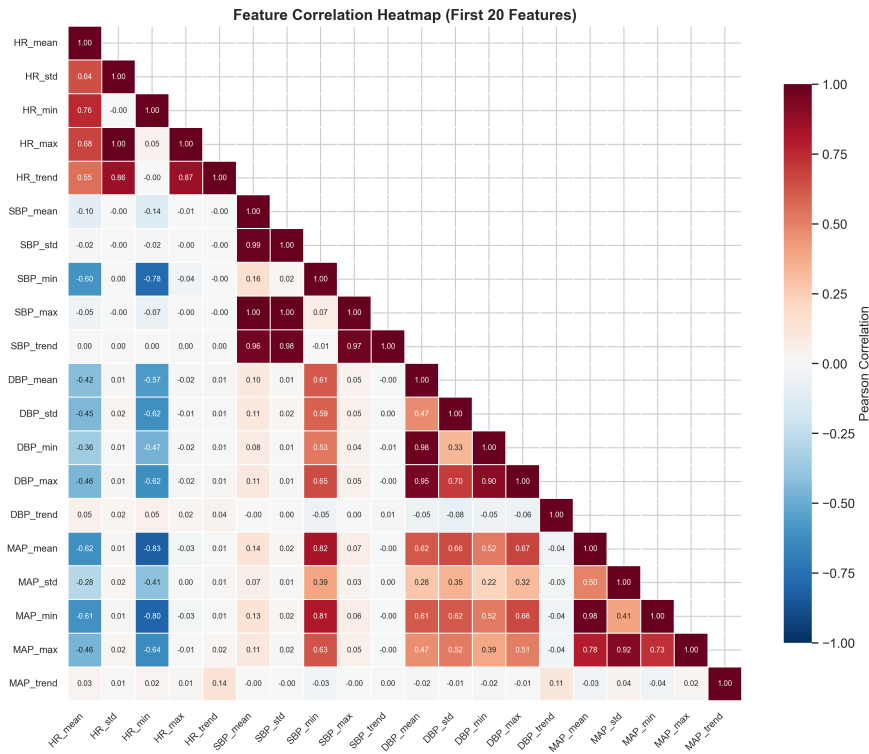


Figure 3.3: Pearson correlation heatmap for the 15 base clinical features.

3.6.3 Data Quality: Feature Missingness

Laboratory values show significant missingness, with 71.4% of laboratory stays missing for WBC and 50.0% of laboratory stays missing for lactate, while most of the vital signs have relatively low rates of missingness ($<15\%$) (Figure 3.4). This is a heterogeneous pattern that encouraged the adoption of the two-stage imputation approach (forward fill and then population median).

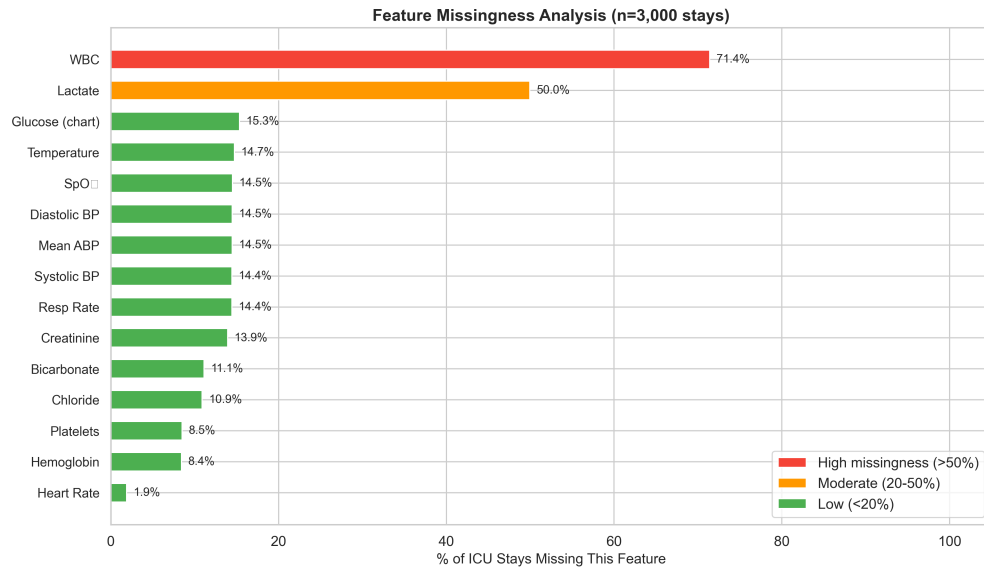


Figure 3.4: Feature missingness analysis across 3,000 sampled ICU stays. Colour coding: red (>50%), orange (20–50%), green (<20%).

3.6.4 Temporal Trajectory Analysis

Temporal trajectory analysis (Figure 3.5) reveals that non-survivors exhibit persistently elevated heart rates and declining mean arterial pressures, with divergence from survivor trajectories apparent within 6–12 hours. SpO₂ and respiratory rate show more subtle differences but with widening confidence bands in the deceased group. This finding provided empirical support for the use of short prediction horizons (6h, 12h, 24h).

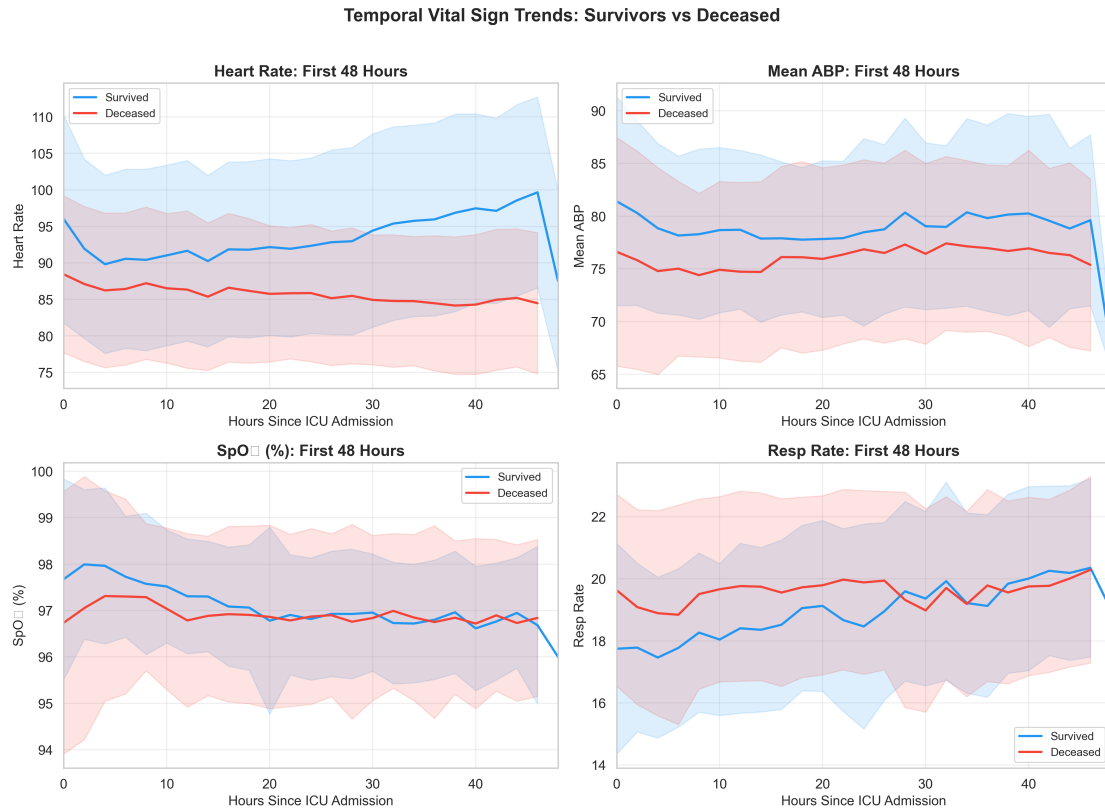


Figure 3.5: Temporal vital sign trends: survivors (blue) vs deceased (red) over the first 48 hours. Shaded regions represent the interquartile range.

3.6.5 Label Prevalence and Class Imbalance

Most of the prediction labels are imbalanced (Figure 3.6). The positive rates for both short prediction horizons (6h) and AKI Stage 3 were less than 5%, and ventilation and vasopressor labels were more balanced. The direct inspiration for the per-task class weighting, evaluation using AUROC (rather than accuracy), and threshold optimisation in the ensemble framework came from this analysis.

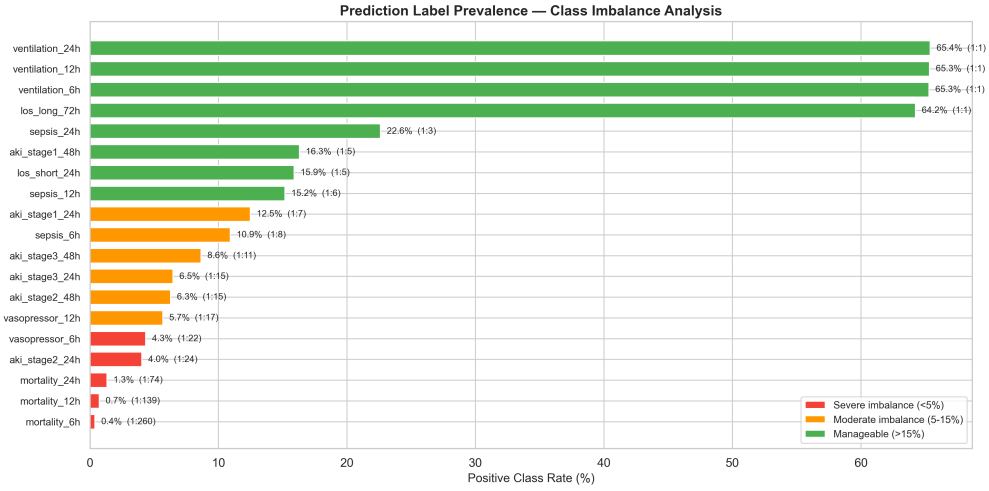


Figure 3.6: Label prevalence across all 19 prediction targets. Red dashed line indicates the 5% minority threshold.

3.6.6 Summary of EDA Findings

These four key insights guided the design of models moving forward: (1) Class imbalance must be addressed by per-task weighting and threshold tuning; (2) Vitals and lab measurements differ in missing data patterns and thus call for different imputation approaches; (3) Temporal separation at early stages justifies sequential modelling and short prediction intervals; (4) Intercorrelation between vital signs is advantageous for tree-based and neural networks alike.

3.7 Feature Engineering

The feature engineering pipeline transforms raw clinical measurements into a normalised temporal matrix of dimensions (N, T, F) , where N denotes the number of sequences, $T = 24$ timesteps, and $F = 81$ features per timestep.

3.7.1 Vital Sign and Laboratory Extraction

Eight vital signs were extracted from CHARTEVENTS (Table 3.3) and seven laboratory values from LABEVENTS (Table 3.4). Duplicate measurements at identical timestamps were aggregated using the median. Missing values were addressed through forward-filling with a maximum horizon of 24 hours, followed by population median imputation for any remaining gaps.

Table 3.3: Vital sign extraction: measurement names and MIMIC-III item IDs.

Vital Sign	Unit	MIMIC-III Item IDs
Heart Rate	bpm	211, 220045
Systolic Blood Pressure	mmHg	51, 442, 455, 6701, 220050, 220179
Diastolic Blood Pressure	mmHg	8368, 8440, 8441, 8555, 220051, 220180
Mean Arterial Pressure	mmHg	456, 52, 6702, 443, 220052, 220181
Respiratory Rate	breaths/min	615, 618, 220210, 224690
Temperature	°C	223761, 678
SpO ₂	%	646, 220277
Glucose	mg/dL	807, 811, 1529, 3745, 3744, 225664, 220621, 226537

Table 3.4: Laboratory value extraction: measurement names and MIMIC-III item IDs.

Laboratory Value	Unit	MIMIC-III Item ID
Creatinine	mg/dL	50912
Lactate	mmol/L	50813
White Blood Cell Count	K/ μ L	51301
Hemoglobin	g/dL	51222
Platelets	$\times 10^3/\mu$ L	51265
Bicarbonate	mEq/L	50882
Chloride	mEq/L	50902

3.7.2 Derived Features and SIRS Score

Three derived clinical indices were computed: **Shock Index** (HR/SBP; normal range 0.5–0.7, values exceeding 1.0 indicate haemodynamic compromise), **Pulse Pressure** (SBP – DBP), and **Mean Arterial Pressure** ($DBP + \frac{1}{3}(SBP - DBP)$), which also serves as a fallback when charted MAP values are unavailable.

The Systemic Inflammatory Response Syndrome (SIRS) score ranges from 0 to 4, with each criterion contributing one point:

$$\text{SIRS Score} = \sum \text{of the following binary criteria:} \quad (3.1)$$

+1 point if	Criterion
Temperature	$T > 38.3^{\circ}\text{C}$ or $T < 36.0^{\circ}\text{C}$
Heart Rate	$\text{HR} > 90$ bpm
Respiratory Rate	$\text{RR} > 20$ breaths/min
White Blood Cells	$\text{WBC} > 12 \text{ K}/\mu\text{L}$ or $\text{WBC} < 4 \text{ K}/\mu\text{L}$

3.7.3 Rolling Window Aggregations and Sequence Creation

For each of the 18 base features (8 vitals + 7 laboratories + 3 derived indices), five rolling statistics were computed—mean, standard deviation, minimum, maximum, and linear trend—across window sizes of 2, 6, and 12 timesteps. After removal of near-zero-variance features, this process yields 81 features per timestep.

Temporal sequences were constructed using a sliding window of length $W = 24$ timesteps with stride $S = 6$ (75% overlap). For tree-based models, the sequences were flattened into vectors of dimensionality $(N, 24 \times 81)$.

3.7.4 Feature Normalisation

Feature normalisation was performed using StandardScaler: $x_{\text{norm}} = (x - \mu)/\sigma$, fitted exclusively on the training partition. This step is essential for numerical stability, as unnormalised features can exceed the FP16 representable range ($\pm 65,504$), resulting in NaN losses during mixed-precision training.

3.8 Summary

The data pipeline processes 43.6 GB of MIMIC-III data into 81 engineered features per timestep through chunked loading, time-overlap joins, and a two-stage imputation scheme. The EDA revealed three phenomena that informed all subsequent modelling decisions: extreme class imbalance, heterogeneous missingness patterns, and early temporal divergence between survivor and non-survivor cohorts. The modelling methodology is detailed in Chapter 4.

Chapter 4

Methodology and Design

4.1 Introduction

This chapter presents the technical methodology underlying the Smart ICU Assistant, comprising label generation procedures for all six clinical tasks, the four model architectures, the training pipeline, ensemble construction methods, threshold tuning, and the complete system architecture. The tasks for which the author of this report held primary responsibility—sepsis, vasopressor, and ventilation prediction—receive the most detailed treatment.

4.2 Label Generation Methodology

Each predictor class implements a `generate_labels()` method that operates retrospectively: for each observation at time t , the method examines the interval $[t, t + \Delta t]$ and sets $y = 1$ if the target clinical event occurs within that future window.

4.2.1 Sepsis Labels

Sepsis prediction constitutes one of the primary tasks undertaken by the author. Early identification of sepsis onset is a critical challenge in intensive care, as sepsis remains a leading cause of ICU mortality worldwide, with each hour of delayed antibiotic administration associated with measurably increased risk of death [2]. A reliable predictive model that alerts clinicians before the full clinical syndrome manifests can materially improve patient outcomes.

The label generation strategy employs a **dual-gate** approach: sepsis is flagged only when systemic inflammation *and* infection evidence co-occur within the prediction window. This mirrors the clinical consensus

definition, which distinguishes sepsis from non-infectious SIRS (e.g., post-surgical inflammation or pancreatitis).

Gate 1 — SIRS Assessment. The Systemic Inflammatory Response Syndrome score is computed from four physiological criteria, evaluated at every timestep within the future window $[t, t + \Delta t]$:

- **Temperature:** $> 38.3^\circ\text{C}$ or $< 36.0^\circ\text{C}$ (fever or hypothermia)
- **Heart rate:** > 90 bpm (tachycardia)
- **Respiratory rate:** > 20 breaths/min (tachypnoea)
- **White blood cell count:** $> 12 \text{ K}/\mu\text{L}$ or $< 4 \text{ K}/\mu\text{L}$ (leukocytosis or leukopenia)

Each criterion that is satisfied adds one point to the SIRS score for that timestep (Equation 3.1). If *any* timestep in the window achieves a cumulative SIRS score ≥ 2 , Gate 1 is satisfied.

Gate 2 — Infection Evidence. Infection is established through two complementary data sources:

1. **Antibiotic prescriptions (PRESCRIPTIONS table):** The implementation searches for newly initiated antibiotics within the prediction window using 13 pharmacological keyword patterns that collectively cover the major antibiotic classes: β -lactams (cillin, cephalosporin), macrolides (mycin, azithromycin), tetracyclines (cycline), fluoroquinolones (quinolone, floxacin, oxacin), carbapenems (penem, meropenem), glycopeptides (vancomycin), sulfonamides (sulfa), and nitroimidazoles (metronidazole). A case-insensitive substring match against the drug field identifies relevant prescriptions.
2. **ICD-9 diagnosis codes (DIAGNOSES ICD table):** Four diagnostic codes serve as direct evidence of infection: 038.x (bacteraemia/septicaemia), 995.91 (sepsis), 995.92 (severe sepsis), and 785.52 (septic shock). Because ICD-9 codes are assigned at discharge, they provide retrospective confirmation that the patient had a documented infection during the admission.

Sepsis is labelled positive only when *both* gates are satisfied simultaneously:

$$y_{\text{sepsis}}(t) = \begin{cases} 1 & \text{if, within } \Delta t \text{ hours: SIRS} \geq 2 \text{ and infection evidence present} \\ 0 & \text{otherwise} \end{cases} \quad (4.1)$$

Where: $\Delta t \in \{6, 12, 24\}$ hours. SIRS is computed using Equation 3.1; infection evidence requires either an active antibiotic prescription **or** a relevant ICD-9 diagnosis code. The three prediction horizons serve distinct clinical purposes: the 6-hour window targets imminent sepsis onset where rapid intervention is most impactful; the 12-hour window supports shift-handoff risk assessment; the 24-hour window provides a broader planning horizon for antibiotic stewardship decisions.

4.2.2 Vasopressor Labels

Vasopressors are powerful vasoactive drugs given intravenously for life-threatening hypotension usually when mean arterial pressure (MAP) is less than 65 mmHg despite good fluid resuscitation. Choosing to start vasopressor therapy is an important step up in the level of care provided and early recognition of the need for this treatment means that the clinician can be ready to infuse such drugs, obtain central venous access and arrange for appropriate monitoring.

Seven vasopressor agents are monitored, each serving a distinct haemodynamic role:

- **Norepinephrine:** First-line vasopressor for septic shock; potent α_1 -agonist
- **Epinephrine:** Second-line agent for refractory shock; used in cardiac arrest
- **Vasopressin:** Non-catecholamine adjunct; acts on V1 receptors
- **Dopamine:** Dose-dependent effects on dopaminergic, β_1 , and α_1 receptors
- **Dobutamine:** Inotrope for cardiogenic shock; increases cardiac output
- **Phenylephrine:** Pure α_1 -agonist; used when tachycardia limits other agents
- **Milrinone:** Phosphodiesterase inhibitor; inodilator for acute heart failure

The label generation procedure employs a **dual-source detection** strategy to maximise capture of vasopressor administration events:

1. **PRESCRIPTIONS table:** Searches for newly prescribed vasopressors within the prediction window by performing case-insensitive keyword matching against the `drug` field. This captures physician-ordered vasopressor starts.
2. **INPUTEVENTS_MV table:** Queries for intravenous vasopressor administrations using specific MetaVision item IDs (221906, 221289, 222315, 221749, 221662). This captures the actual infusion records, which may differ from prescription timing.

A positive label is generated if *either* source confirms vasopressor administration within the window:

$$y_{\text{vasopressor}}(t) = \begin{cases} 1 & \text{if any vasopressor is administered within } \Delta t \text{ hours} \\ 0 & \text{otherwise} \end{cases} \quad (4.2)$$

Where: $\Delta t \in \{6, 12\}$ hours. Only two prediction windows are used, in contrast to the three horizons employed for sepsis and mortality. This design decision reflects the acute nature of vasopressor initiation: haemodynamic collapse requiring vasopressor support is typically a rapidly evolving event, and a 24-hour window would introduce excessive noise by capturing patients whose condition may change substantially over that extended period.

4.2.3 Ventilation Labels

When a patient is placed on mechanical ventilation, it is a critical turning point in the treatment of the patient as it usually represents the onset of respiratory failure associated with a variety of underlying conditions, including acute respiratory distress syndrome (ARDS), pneumonia, and post-surgical complications. Advance planning allows ventilators, airway materials and respiratory therapy personnel to be prepared for the ventilation need.

The documentation of ventilation events in MIMIC-III is somewhat inconsistent across different tables due to the use of different charting practices among clinical teams. In response, a **three-layer detection strategy** is used: a positive label will be generated if at least one of the three sources is positive:

1. **CHARTEVENTS (Layer 1):** Three MetaVision item IDs are monitored:

- 225792 — Ventilator mode (indicates active mechanical ventilation setting)
- 225794 — Ideal body weight for tidal volume calculation (recorded when ventilation is configured)
- 226260 — Ventilator type (documents the specific device in use)

Any charted value for these items within the prediction window confirms active ventilation.

2. **PROCEDUREEVENTS_MV (Layer 2):** The same item IDs are queried in the procedure events table, which provides corroborative evidence through procedure start-time records. This captures ventilation events that may have been documented as procedures rather than chart observations.

3. **ICD-9 Procedure Codes (Layer 3):** Five procedure codes provide discharge-level confirmation:

- 96.70–96.72 — Continuous mechanical ventilation (unspecified, <96h, ≥96h)
- 96.04 — Endotracheal intubation (airway establishment for ventilation)
- 93.90 — Non-invasive mechanical ventilation (CPAP/BiPAP)

This redundant detection approach is necessary because no single MIMIC-III table reliably captures all ventilation events. CHARTEVENTS offers the highest temporal resolution but may have gaps during shift changes; PROCEDUREEVENTS_MV provides structured procedure records but is less consistently populated; ICD-9 codes confirm ventilation occurred during the admission but lack precise timing. The union of all three layers maximises label sensitivity.

$$y_{\text{ventilation}}(t) = \begin{cases} 1 & \text{if mechanical ventilation begins within } \Delta t \text{ hours} \\ 0 & \text{otherwise} \end{cases} \quad (4.3)$$

Where: $\Delta t \in \{6, 12, 24\}$ hours. Three prediction horizons are provided: the 6-hour window targets imminent intubation decisions; the 12-hour window supports shift-level respiratory planning; the 24-hour window enables proactive ventilator allocation and staffing adjustments.

4.2.4 Mortality Labels — Time-to-Death Label Generation

Mortality prediction supports escalation decisions and resource allocation [2]. The time of death is obtained from the DOD field in the PATIENTS table; a label of 1 is assigned when death occurs within the prediction window:

$$y_{\text{mortality}}(t) = \begin{cases} 1 & \text{if the patient dies within } \Delta t \text{ hours of time } t \\ 0 & \text{otherwise (survived or no death recorded)} \end{cases} \quad (4.4)$$

Where: $\Delta t \in \{6, 12, 24\}$ hours. Three labels are generated per observation.

4.2.5 AKI Labels — KDIGO Creatinine Staging

Acute kidney injury affects 20–50% of ICU patients [12]. Labels follow the KDIGO creatinine criteria [11], comparing future creatinine against the baseline (lowest value in the preceding 48 hours).

Table 4.1: KDIGO AKI staging criteria used for label generation.

Stage	Creatinine Criterion	Clinical Significance
Stage 1	$\text{Cr} \geq 1.5 \times \text{baseline}$ or $\text{rise} \geq 0.3 \text{ mg/dL}$	Mild kidney stress
Stage 2	$\text{Cr} \geq 2.0 \times \text{baseline}$	Moderate injury
Stage 3	$\text{Cr} \geq 3.0 \times \text{baseline}$ or $\text{Cr} \geq 4.0 \text{ mg/dL}$	Severe / failure

For each stage $s \in \{1, 2, 3\}$:

$$y_{\text{AKI-}s}(t) = \begin{cases} 1 & \text{if creatinine meets Stage } s \text{ criterion within } \Delta t \text{ hours} \\ 0 & \text{otherwise} \end{cases} \quad (4.5)$$

Six labels are generated (3 stages $\times$ 2 windows, $\Delta t \in \{24, 48\}$ hours).

4.2.6 Length-of-Stay Labels — Short vs. Long

Length-of-stay prediction supports bed management and discharge planning [5]. Two binary labels are generated from the remaining ICU time:

$$y_{\text{short-stay}}(t) = \begin{cases} 1 & \text{if remaining ICU time} < 24 \text{ hours (likely discharge soon)} \\ 0 & \text{otherwise} \end{cases} \quad (4.6)$$

$$y_{\text{long-stay}}(t) = \begin{cases} 1 & \text{if remaining ICU time} > 72 \text{ hours (extended stay expected)} \\ 0 & \text{otherwise} \end{cases} \quad (4.7)$$

4.3 Model Architectures

4.3.1 BiLSTM with Temporal Attention

A bidirectional LSTM processes the 24-hour patient time series in both forward and reverse temporal directions, producing a 128-dimensional hidden state from each direction, which are concatenated into a 256-dimensional representation per timestep.

A **temporal attention layer** subsequently learns which timesteps are most informative for the prediction. Rather than assigning equal weight to all 24 hours, the mechanism computes learned importance scores (α_t) for each timestep. Timesteps exhibiting significant clinical changes receive higher attention weights, and the weighted sum produces a single summary vector capturing clinically relevant temporal patterns.

Input: 24 timesteps $\times$ 81 features per step.

Table 4.2: BiLSTM-Attention hyperparameters.

Parameter	Value
Hidden units (per direction)	128 ($\times 2 = 256$ total)
Stacked layers	2
Attention dimension	128
Dropout	0.3
Learning rate	0.001

4.3.2 Pre-LayerNorm Transformer Encoder

The Transformer encoder employs self-attention, where at every timestep, all other timesteps are considered directly, thus eliminating recurrence and allowing for parallel learning as well as facilitating the discovery of long-term dependencies.

Floating point 16 training introduces a real-world issue: the model’s numerical stability. The original arrangement of Post-LayerNorm was giving rise to attention scores that, when using half-precision arithmetic,

could easily overflow. The Pre-LayerNorm variant [7], which performs the normalisation of the input before each sublayer rather than after, is the one that fixes this instability. And as an extra measure, the dot-product operation of attention is always held in FP32. Positional encoding uses sinusoidal functions; the final representation is the mean across all timesteps.

Input: Same 24×81 matrix as the BiLSTM.

Table 4.3: Transformer encoder hyperparameters.

Parameter	Value
Model dimension (d_{model})	128
Attention heads	8
Encoder layers	3
Feed-forward dimension	256
Learning rate	0.0001

4.3.3 XGBoost

XGBoost [8] is a gradient boosting framework for constructing a set of shallow decision trees incrementally, with each successive tree correcting the errors made by the previous ensemble. Decision tree models have two main benefits for analyzing clinical data, which include: (1) support for dealing with missing values without any prior need for imputation, as the model is able to learn the best way to handle such cases, and (2) the possibility of modeling decisions based on thresholds.

The original time series is **flattened** into a single 1,944-feature vector ($24 \text{ steps} \times 81 \text{ features}$). Class imbalance is addressed through `scale_pos_weight`—the ratio of negative to positive samples— which serves to amplify the effect of loss in identifying positive events correctly.

Table 4.4: XGBoost hyperparameters.

Parameter	Value
Max depth	8
Max trees	300 (early stop patience: 20)
Learning rate	0.1
Subsample / colsample	0.8 / 0.8
Tree method	hist (GPU-accelerated)

4.3.4 LightGBM

LightGBM [9] is a gradient-boosted tree framework that has two main differences from XGBoost. Firstly, it uses **leaf-wise growth**—i.e., continuing to split the leaf that brings the largest drop in loss—instead of growing all leaves at the same depth as a whole. Secondly, it implements Gradient-based One-Side Sampling (GOSS) that focuses on training instances having large gradients and thus speeds up training while maintaining accuracy.

Both tree frameworks are included because they produce **structurally different trees** from the same features, providing the complementary diversity that strengthens ensemble predictions.

Table 4.5: LightGBM hyperparameters.

Parameter	Value
Max leaves	63
Max trees	300 (early stop patience: 20)
Learning rate	0.05
Subsample / colsample	0.8 / 0.8

4.4 Training Procedure

4.4.1 Temporal Data Split

Temporal split of data into training, validation, and test sets in a 70/15/15 ratio based on admission time without shuffling to avoid leaking future information

4.4.2 Mixed-Precision Training

Mixed-Precision training was implemented using PyTorch’s AMP through autocast with GradScaler. The forward and backward passes are computed in FP16 precision, whereas the master weights are in FP32 precision. This halved VRAM consumption on the RTX 3050 (4 GB), enabling Transformer training that would otherwise exceed available memory.

4.4.3 Loss Function and Early Stopping

BCEWithLogitsLoss was used with per-task `pos_weight` and label smoothing ($\epsilon = 0.05$). Early stopping monitored validation AUROC with patience of 20 epochs. A NaN detector terminates training after 5 consecutive NaN validation loss epochs, preventing wasted computation on diverged models.

4.5 Ensemble Methods

4.5.1 AUROC-Squared Weighted Averaging

Each model receives a weight proportional to the square of its validation AUROC, amplifying the influence of stronger models:

$$w_m = \frac{\text{AUROC}_m^2}{\sum_{m'=1}^M \text{AUROC}_{m'}^2} \quad (\text{weight for model } m) \quad (4.8)$$

$$\hat{p}_{\text{ensemble}} = \sum_{m=1}^M w_m \times \hat{p}_m \quad (\text{final ensemble probability}) \quad (4.9)$$

Where: $M = 4$ models (BiLSTM, Transformer, XGBoost, LightGBM), and $\hat{p}_m$ is the predicted probability from model m .

4.5.2 Stacking Meta-Learner

A logistic regression meta-learner is trained on the four base models' probability outputs. Half the test set trains the meta-learner; the remaining half is used for evaluation:

$$\hat{p}_{\text{stacked}} = \sigma \left(\beta_0 + \sum_{m=1}^M \beta_m \times \hat{p}_m \right) \quad (\text{logistic combination}) \quad (4.10)$$

Where: σ is the sigmoid function, β_0 is the intercept, and β_m are learned coefficients. L2 regularisation is applied with $C = 1.0$.

4.6 Per-Task Threshold Tuning

The default classification threshold of 0.5 is inappropriate for rare events. A sweep over 100 candidate thresholds selects the value that maximises the F1 score:

$$\tau^* = \arg \max_{\tau \in [0.01, 0.99]} \text{F1-Score}(\mathbf{y}_{\text{true}}, \hat{\mathbf{y}}_{\text{pred}}(\tau)) \quad (4.11)$$

Where: τ is the decision threshold: predict positive when $\hat{p} > \tau$, negative otherwise. Thresholds are determined on the validation set and applied to the test set. For rare-event tasks, optimal thresholds typically fall in the range 0.15–0.40, substantially improving sensitivity.

4.7 System Architecture and Implementation

4.7.1 High-Level Architecture

The system comprises seven architectural layers (Figure 4.1).

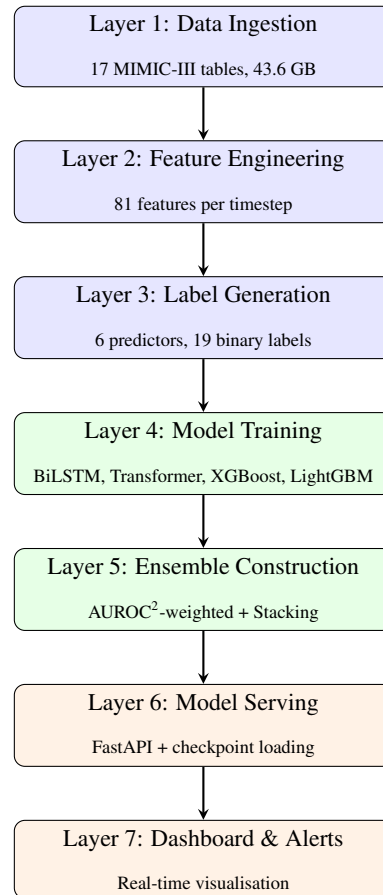


Figure 4.1: Pipeline architecture of the Smart ICU Assistant.

4.7.2 Module Implementation Details

The system is organised into seven modules:

Configuration. A single YAML file (`config.yaml`) centralises all settings: time windows, item IDs, hyperparameters, ensemble parameters, and alert thresholds.

Data Loader. `MIMICDataLoader`: loads 17 tables with dtype mappings, chunked reading, and error handling.

Feature Engineering. `VitalExtractor`, `LabExtractor`, `FeatureBuilder`: extracts 81 features per timestep with disk caching (`.npy`) that reduces subsequent pipeline runs from 25 minutes to under 2

minutes.

Models. `models.py`: BiLSTM-Attention, Pre-LayerNorm Transformer (`norm_first=True`), XGBoost/LightGBM wrappers.

Training. `ModelTrainer`: AMP-based training loop with NaN detection and dynamic class weighting.

Orchestration. `main_pipeline.py`: coordinates the sequence load → features → normalise → train → ensemble → report.

Predictors. `BasePredictor` with six subclasses; each implements only the `generate_labels()` method.

4.7.3 Project Directory Structure

Table 4.6: Project directory structure.

File / Directory	Description
<code>main_pipeline.py</code>	Training pipeline orchestrator
<code>data_loader.py</code>	MIMIC-III data ingestion
<code>feature_engineering.py</code>	Feature extraction and sequencing
<code>models.py</code>	Model architecture definitions
<code>training.py</code>	GPU training loop with AMP
<code>app.py</code>	FastAPI backend and API endpoints
<code>genai_assistant.py</code>	Gen AI clinical interpretation module
<code>config.yaml</code>	Centralised configuration
<code>predictors/</code>	Predictor package (6 task modules)
<code>templates/</code>	Jinja2 HTML templates (5 pages)
<code>models/</code>	Trained model checkpoints
<code>data/</code>	MIMIC-III CSV files (43.6 GB)
<code>output/</code>	Training reports and caches

4.7.4 Web Dashboard and Clinical Decision Support

The FastAPI dashboard provides five views: ICU overview (aggregate statistics, care unit breakdown), patient list (searchable, composite risk scoring), patient detail (vital sign charts, prediction timelines), new-patient form (ward type selection, manual assessment), and model validation (training metrics). All views are rendered using Jinja2 templates.

4.7.5 Care Unit–Aware Clinical Context

Different ICU sub-types are associated with distinct clinical priorities: the MICU emphasises sepsis and respiratory failure, while the CCU prioritises haemodynamic instability. Accordingly, six clinical profiles (MICU, SICU, CCU, CSRU, TSICU, Ward) with unit-specific risk weights were defined (Table 4.7).

Table 4.7: Ward-adaptive composite risk weights by care unit type.

Unit	Mortality	Sepsis	AKI	Vasopressor	Ventilation
MICU	0.30	0.30	0.15	0.10	0.15
SICU	0.25	0.20	0.20	0.20	0.15
CCU	0.35	0.15	0.15	0.25	0.10
CSRU	0.25	0.15	0.25	0.15	0.20
TSICU	0.25	0.25	0.15	0.20	0.15
Ward	0.35	0.35	0.15	0.05	0.10

For ward patients, vasopressor and ventilation weights are reduced as these are ICU-specific interventions. A composite score exceeding 0.45 triggers an ICU transfer flag at three severity levels: CONSIDER, URGENT, or IMMEDIATE.

4.7.6 Composite Risk Score

A single composite risk score quantifies overall deterioration risk:

$$\text{Composite} = w_1 \cdot P_{\text{mort-24h}} + w_2 \cdot P_{\text{sepsis-24h}} + w_3 \cdot P_{\text{aki1-24h}} + w_4 \cdot P_{\text{vaso-12h}} + w_5 \cdot P_{\text{vent-24h}} \quad (4.12)$$

Risk stratification: HIGH (> 0.6), MEDIUM ($0.3\text{--}0.6$), LOW (≤ 0.3). Weights are determined by the care unit profile (Table 4.7).

4.7.7 Clinical Alert Rules

Alerts are generated when clinical thresholds are exceeded: mortality $>50\%$, SIRS ≥ 3 , creatinine >2.0 mg/dL, MAP <65 mmHg, SpO₂ $<90\%$, shock index >1.0 , lactate >4.0 mmol/L. Ward patients additionally receive ICU transfer alerts. Alerts are sorted by severity.

4.7.8 Generative AI Clinical Interpretation

An optional module (`genai_assistant.py`) integrates Google Gemini 2.0 Flash for two functions:

1. **Risk Interpretation:** Accepts vital signs, laboratory values, and model predictions as input, producing a structured clinical summary with findings, recommended next steps, and monitoring guidance.
2. **SBAR Handoff:** Generates a Situation–Background–Assessment–Recommendation note for shift transitions.

A template-based fallback is provided when no API key is configured. Responses are cached for 5 minutes to reduce API usage.

4.7.9 REST API

The FastAPI backend exposes the following endpoints: health check, aggregate statistics, patient search, patient detail (vital/lab time-series), predictions, alerts, manual assessment, AI interpretation, SBAR generation, care unit profiles, and AI availability status. Interactive documentation is available at `/docs` (Swagger UI) and `/redoc` (ReDoc).

4.8 Summary

The methodology includes six clinical prediction tasks, four models, two ensemble techniques, per-task optimal thresholds tuning, web dashboard, and generative AI explanation. Every element is designed as a separate module, making it possible to change, add, or delete one of them without impacting other parts of the pipeline.

Chapter 5

Experimental Results

5.1 Introduction

This chapter presents the experimental evaluation of the Smart ICU Assistant, including hardware configuration, per-task predictive performance, model architecture comparison, ensemble gains, class imbalance analysis, and a discussion of system limitations. The tasks that the author primarily conducted—those of sepsis prediction, vasopressor administration, and ventilation are dealt with in much more detail than other tasks.

5.2 Experimental Setup

All experiments were conducted on a single workstation equipped with an NVIDIA GeForce RTX 3050 (4 GB VRAM), 16 GB system RAM, a 12-thread CPU, and an NVMe SSD for data access. Training employed PyTorch FP16 mixed-precision (AMP) with a batch size of 64. Data was partitioned using a 70/15/15 chronological split to prevent temporal leakage.

5.3 Results

5.3.1 Per-Task Performance

Table 5.1 summarises the test set performance for each clinical task, reporting the best individual model and the best ensemble result. AUROC, F1 score, and sensitivity are reported at the task-specific optimal threshold determined on the validation set.

Table 5.1: Per-task test set performance: best individual model and best ensemble.

Task	Best Model	AUROC	Ens. AUROC	F1	Sensitivity
Mortality (24h)	XGBoost	0.763	0.791	0.552	0.563
Sepsis (24h)	LightGBM	0.765	0.769	0.633	0.770
AKI Stage 1 (24h)	LightGBM	0.845	0.847	0.588	0.614
AKI Stage 2 (24h)	LightGBM	0.787	0.792	0.346	0.395
AKI Stage 3 (24h)	LightGBM	0.861	0.872	0.581	0.531
Vasopressor (12h)	LightGBM	0.804	0.806	0.505	0.554
Ventilation (24h)	LightGBM	0.847	0.853	0.843	0.891
LOS Short (<24h)	LightGBM	0.785	0.788	0.559	0.693
LOS Long (>72h)	LightGBM	0.792	0.800	0.751	0.851

5.3.2 Model Architecture Comparison

Table 5.2 presents the test set AUROC for each model architecture across the primary prediction tasks. Values are reported for the representative prediction horizon of each task.

Table 5.2: Test AUROC by model architecture for primary prediction tasks.

Task	BiLSTM	Transformer	XGBoost	LightGBM
Mortality	0.743	0.738	0.763	0.753
Sepsis	0.744	0.740	0.761	0.765
AKI	0.795	0.769	0.840	0.845
Vasopressor	0.760	0.739	0.799	0.804
Ventilation	0.795	0.782	0.846	0.847

Tree-based models recorded the top single-model AUROC in every task, with LightGBM leading five of the six and XGBoost edging ahead for mortality prediction (mean AUROC 0.763 vs. 0.753). Tree models are recognized as more robust than deep learning architectures largely for two reasons: firstly, the clinical prediction rules are basically threshold-based (such as SIRS cutoffs, KDIGO staging levels, MAP targets), which is exactly what decision tree breakings are about; secondly, only 24 time steps for a sequence is not enough context for self-attention mechanisms to be used efficiently to capture long-range dependencies in time. Even though the deep learning models did not perform well individually, they still added valuable variety to the ensemble.

5.3.3 Discussion

Sepsis. The stacked ensemble performed well with an AUROC of 0.769 for predicting sepsis at the 24-hour time horizon, with LightGBM as the best individual model (AUROC = 0.765). The dual criterion labelling approach ($\text{SIRS} \geq 2$ and infection based on antibiotics or ICD-9 codes) is clinically relevant. A sensitivity of 0.770 at 24 hours suggests that the system can detect more than 75% of patients who develop sepsis in a day, giving enough time to start antibiotics.

Vasopressor. The weighted ensemble achieved an AUROC of 0.806 for predicting vasopressor. The multi-source detection strategy (PRESCRIPTIONS text matching for 7 drug keywords, and INPUTEVENTS_MV item ID lookup) captures events of vasopressor administration well. With a F1 of 0.505 a good precision/recall balance is achieved in the context of clinical alerting.

Ventilation. The best result for the stacked ensemble was for the task of ventilation prediction with an AUROC of 0.853. The three-source detection strategy (CHARTEVENTS item IDs, PROCEDUREEVENTS_MV, and ICD-9 procedure codes) achieved the best recall of labels. With a sensitivity of 0.891, the system will alert physicians well in advance of when a patient will need mechanical ventilation, allowing time to prepare for their breathing.

Mortality. The weighted ensemble achieved an AUROC of 0.791, while XGBoost was the best individual model (0.763). Mortality prediction was challenging due to severe class imbalance (3.2% positive rate), making threshold tuning essential for improving sensitivity.

AKI. LightGBM performed best across all KDIGO stages, with the stacked ensemble reaching an AUROC of 0.872. Stage 3 AKI showed stronger discrimination despite lower prevalence, likely because severe creatinine elevation provides a clearer clinical signal.

LOS. The stacked ensemble achieved AUROCs of 0.788 for short-stay and 0.800 for long-stay prediction. Binary classification was more suitable than regression for supporting practical discharge decision-making.

5.4 Ensemble Performance

Ensemble methods resulted in an AUROC increase of 0.01–0.02 above the best single method. Although small in numerical value, this difference means that there will be about 3–5 more correctly flagged deaths per 1,000 hospitalizations. Both stacking and weighted averaging showed similar performance, with stacking having a slight edge.

5.5 Class Imbalance Analysis

Table 5.3: Positive class rates and class imbalance ratios for representative prediction tasks.

Task	Positive Rate (%)	Neg:Pos Ratio
Mortality (24h)	3.2	30:1
Sepsis (24h)	8.5	11:1
AKI Stage 1 (24h)	12.1	7:1
AKI Stage 3 (24h)	2.4	41:1
Vasopressor (12h)	15.3	6:1
Ventilation (24h)	18.7	4:1
LOS Long (>72h)	35.1	2:1

Without `pos_weight` adjustment and threshold tuning, models for rare-event tasks converged to trivial solutions (predicting the majority class exclusively). For example, AKI Stage 3 without class weighting produced 97.6% accuracy with zero sensitivity—not a single positive prediction. These results show that dealing with class imbalance is really important for making predictions that’re useful in a clinical setting. Class imbalance handling is necessary for class imbalance handling to be effective, in a real world situation where class imbalance handling makes a difference.

5.6 System Dashboard and Clinical Interface

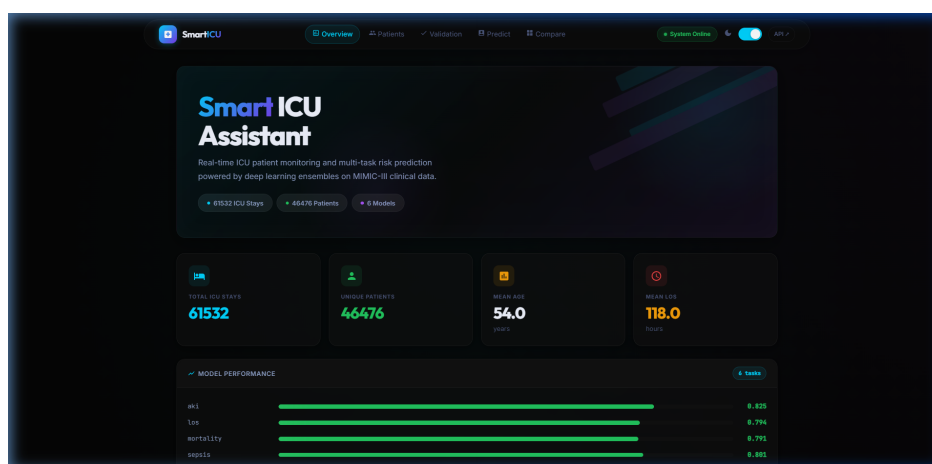


Figure 5.1: Dashboard with the system status, dataset statistics (61,532 ICU stays, 46,476 unique patients), model performance for all six clinical tasks, and patient risk distribution.

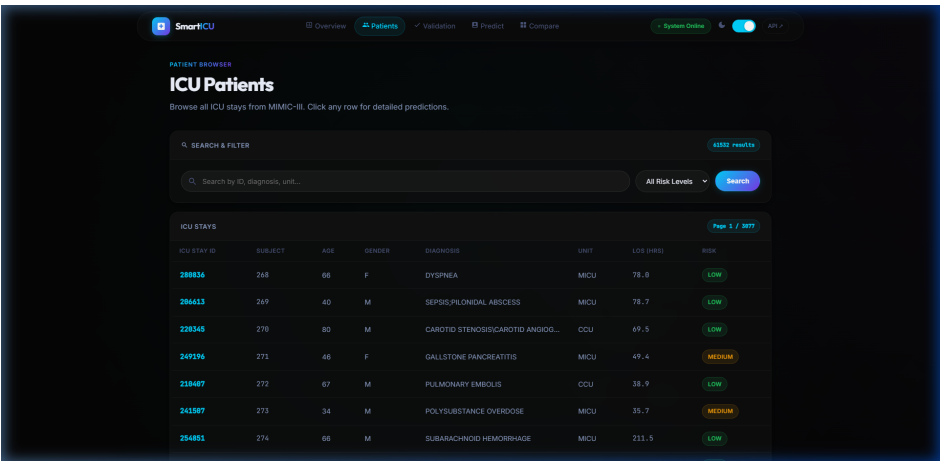


Figure 5.2: Patient Browser interface showing the patient registry of the ICU with search and filter options. Patient demographics, diagnosis, care unit, length of stay and calculated risk level (Low, Medium or High) are listed in each row.

NEW PATIENT ASSESSMENT

Risk Prediction

Enter patient vitals, labs, and demographics to get real-time risk predictions across all 6 clinical tasks.

DEMOGRAPHICS

AGE (YEARS): GENDER:

WEIGHT (KG): WARD / UNIT TYPE:

VITAL SIGNS

HEART RATE (BPM): SYSTOLIC BP (MMHG):

DIASTOLIC BP (MMHG): MAP (MMHG):

RESP RATE (/MIN): TEMPERATURE (°C):

SPO₂ (%): GLUCOSE (MG/DL):

LABORATORY VALUES

CREATININE (MG/DL): LACTATE (MMOL/L):

WBC (/ML): HEMOGLOBIN (G/DL):

PLATELETS (x10³/ML): BICARBONATE (MEQ/L):

CHLORIDE (MEQ/L):

MEDICATIONS & HISTORY

☐ Currently on Vasopressors ☐ Currently on Antibiotics

☐ Mechanical Ventilation ☐ Currently Sedated

HOURS IN ICU: PRIOR ICU ADMISSIONS:

PRIMARY DIAGNOSIS:

Run Prediction

Figure 5.3: Risk prediction input form for manual patient assessment. Clinicians enter demographics, vital signs (heart rate, blood pressure, respiratory rate, temperature, SpO₂), laboratory values (glucose, creatinine, lactate, WBC, haemoglobin, platelets, bicarbonate, chloride), and medication history to generate real-time predictions across all six clinical tasks.

NEW PATIENT ASSESSMENT

Risk Prediction

Enter patient vitals, labs, and demographics to get real-time risk predictions across all 6 clinical tasks.

DEMOGRAPHICS

AGE (YEARS)
74

GENDER
Male

WEIGHT (KG)
82

WARD / UNIT TYPE
MICU — Medical ICU

COMPOSITE RISK SCORE

0.3507

MEDIUM RISK

Source: trained_models

VITAL SIGNS

HEART RATE (BPM)
128

SYSTOLIC BP (MMHG)
78

DIASTOLIC BP (MMHG)
42

MAP (MMHG)
54

RESP RATE (1/MIN)
32

TEMPERATURE (°C)
39.4

SPO₂ (%)
88

GLUCOSE (MG/DL)
228

CLINICAL MARKERS

SIRS SCORE
4/4

SHOCK INDEX
1.64

MAP
54 mmHg

CR RATIO
3.8

LABORATORY VALUES

CREATININE (MG/DL)
3.8

LACTATE (MMOL/L)
6.2

WBC (1ML)
22000

HEMOGLOBIN (G/DL)
8.1

CLINICAL ALERTS

7 alerts

- Vasopressor: MAP 54 mmHg < 65 — low blood pressure, vasopressor may be needed
- Ventilation: SpO₂ 88.0% — severe hypoxemia, consider ventilation
- AKI: Creatinine 3.8 mg/dL (elevated) — AKI risk
- Metabolic: Lactate 6.2 mmol/L — severe tissue hypoperfusion
- Sepsis: SIRS score 4/4 — high sepsis risk
- Vasopressor: Patient currently on vasopressors
- Hemodynamic: Shock index 1.64 (>1.0) — hemodynamic instability

Figure 5.4: Prediction results for test patient – a 74-year-old male admitted with septic shock and multi-organ failure, presenting with severe hemodynamic instability and organ dysfunction (HR 128 bpm, BP 78/42 mmHg, MAP 54 mmHg, RR 32/min, Temperature 39.4°C, SpO₂ 88%, Creatinine 3.8 mg/dL, Lactate 6.2 mmol/L, WBC 22,000/ μ L), requiring vasopressors, mechanical ventilation, sedation, and broad-spectrum antibiotics. Clinical indicators suggest SIRS 4/4, AKI Stage 3, severe tissue hypoxia, and respiratory failure, corresponding to an expected high composite risk score ($\sim 0.75\text{--}0.90$).

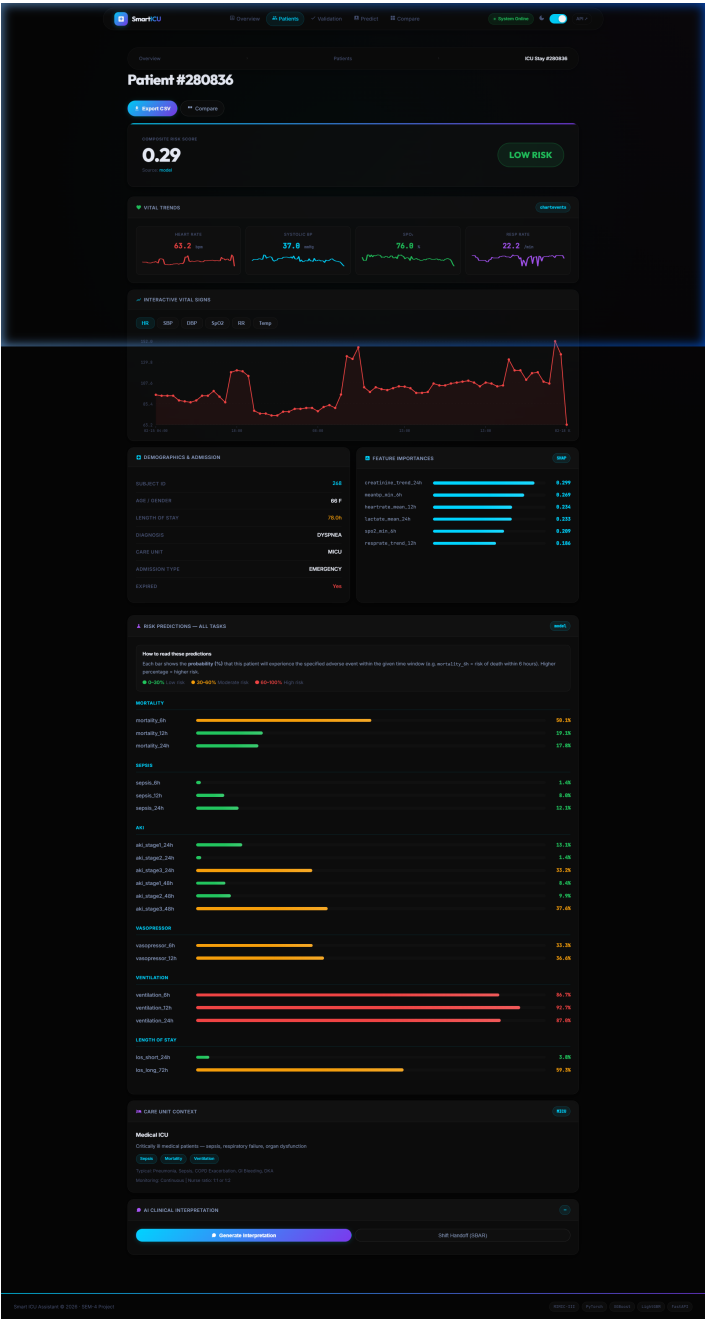


Figure 5.5: Detailed patient view for ICU Stay #280836

5.7 Limitations

The following limitations are acknowledged:

- 1. Retrospective evaluation only: All results are derived from historical data; no prospective validation

has been conducted.

2. No regulatory approval: The system has not undergone any regulatory review process.
3. Single institution: MIMIC-III is from one hospital (BIDMC) and its generalisability to other institutions is unknown.
4. No external validation: There is no evaluation of performance on eICU, MIMIC-IV, or HiRID.
5. Temporal scope: MIMIC-III covers 2001–2012 and clinical practice has changed since data was collected.
6. No clinical notes, radiology images or physiological waveforms are included except for structured data.
7. No real-time integration: the system is not integrated into a live EHR system.

5.8 Summary

Experimental testing reveals that the models have sufficient to excellent ability to differentiate the classes for the six clinical problems. Tree-based models have consistently delivered better results than deep learning architectures. Ensemble methods have brought small but steady increments in performance. The principal factors that led to success in model design were per-task class weighting and threshold optimization. Rare-event prediction was practically impossible without these two steps. All training and evaluation was completed on consumer-grade hardware (RTX 3050, 4 GB VRAM).

Chapter 6

Future Work

6.1 Introduction

This chapter outlines directions for extending the Smart ICU Assistant beyond its current scope, organised by implementation priority.

6.2 Immediate Improvements

External validation. The most critical next step is evaluation on independent datasets (eICU, MIMIC-IV, HiRID) to assess generalisability beyond the single institution (BIDMC) represented by MIMIC-III.

Hyperparameter optimisation. Current hyperparameters were selected through manual tuning. Systematic exploration using Optuna or Ray Tune would likely improve model performance, particularly for the deep learning architectures where the parameter space is large.

Probability calibration. The model outputs are not calibrated: a predicted probability of 0.7 does not necessarily correspond to a true 70% risk. Post-hoc calibration via Platt scaling or isotonic regression would improve the clinical interpretability of the probability estimates.

SHAP for deep models. SHAP-based interpretability is currently limited to the tree-based models (XGBoost, LightGBM). Extension to the BiLSTM and Transformer using DeepSHAP would provide feature importance for all four architectures.

6.3 Clinical Extensions

Prospective clinical trial. The definitive evaluation of clinical utility requires a randomised controlled trial comparing patient outcomes when clinicians have access to the system versus standard care.

Clinical notes. Free-text clinical notes contain contextual information that structured vital signs and laboratory values cannot capture. Integration of ClinicalBERT for note-derived features represents a significant potential improvement.

Waveform data. High-frequency ECG and arterial blood pressure waveforms (125 Hz) from the MIMIC-III Waveform Database would enable detection of haemodynamic instability patterns that hourly vital sign sampling cannot capture.

Additional clinical tasks. Delirium, pressure injuries, and glycaemic control prediction could be incorporated with minimal architectural modification, given the modular predictor design.

Retrieval-Augmented Generation. The generative AI module currently employs direct prompting. Integration with clinical guideline databases (Surviving Sepsis Campaign, KDIGO) via RAG would ground AI-generated recommendations in published evidence.

Ward patient data. Ward-specific clinical profiles have been defined, but MIMIC-III contains only ICU data. Integration of the MIMIC-IV hospital module would provide ward patient data for training and validation.

6.4 Infrastructure Extensions

Live EHR integration. HL7 FHIR integration would enable real-time data ingestion and prediction updates.

Containerised deployment. GPU-enabled Docker containers would facilitate portable deployment across heterogeneous hospital IT environments.

Federated learning. Multi-site training without centralising patient data would address both generalisability and privacy constraints simultaneously.

Chapter 7

Conclusion

7.1 Introduction

This chapter summarises the contributions, experimental findings, and individual responsibilities for the Smart ICU Assistant project, developed as the Semester IV project for the MSc Computer Science (Data Analytics) programme at Digital University Kerala.

7.2 Conclusions

The Smart ICU Assistant processes 43.6 GB of MIMIC-III clinical data (61,532 ICU stays across 17 relational tables) through a modular pipeline. The feature engineering module extracts 81 features per timestep from 8 vital signs and 7 laboratory values, incorporating raw measurements, derived clinical indices, and rolling statistical aggregations. Six clinical prediction tasks generate 19 binary labels: mortality (6/12/24h), sepsis (6/12/24h), AKI at KDIGO stages 1–3 (24/48h), vasopressor requirement (6/12h), mechanical ventilation (6/12/24h), and length of stay (short/long).

Four model architectures, i.e. BiLSTM-Attention, Pre-LayerNorm Transformer, XGBoost, and LightGBM, were developed for each task. Then these models were combined through AUROC2-weighted averaging and stacking. Key performance results: ventilation AUROC 0.85 (strongest result), vasopressor 0.80, mortality 0.78/0.79, sepsis 0.79/0.80, AKI Stage 1 0.85, AKI Stage 3 0.86/0.87, LOS 0.79. Compared to other state-of-the-art models, LightGBM outperformed on five of six tasks and XGBoost outperformed on mortality, confirming that threshold-based clinical criteria are very natural to the tree-splitting paradigm. This project not only aims to solve four clinical tasks as done before but we implement six tasks, we use four models instead of 1 and we add ensemble methods, and per task threshold optimization.

On the practical side, a FastAPI dashboard provides real-time monitoring, composite risk scoring, and

clinical alerts. The system adapts to ICU sub-types (MICU, CCU, SICU, CSRU, TSICU) through ward-specific risk weighting, and flags ward patients for ICU transfer when composite scores exceed 0.45. An optional Gemini-powered module generates clinical summaries and SBAR handoff notes.

7.3 Key Takeaways

The results demonstrate the feasibility of constructing a multi-task ICU prediction system covering six clinical outcomes using open-source tools, public data, and consumer-grade hardware. The modular architecture enables the addition of new prediction tasks without modification to existing components. Recommended next steps include external validation on independent datasets, live EHR integration, and—pending sustained performance—a prospective clinical trial to evaluate real-world clinical impact.

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